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Relativity and the Absolute

IS SPACETIME AN EINSTEINIAN ABSTRACTION OR A PHYSICAL ENTITY?

Some discoveries provide answers to questions. Other discoveries are so deep that they cast questions in a whole new light, showing that previous mysteries were misperceived through lack of knowledge. You could spend a lifetime—in antiquity, some did—wondering what happens when you reach earth's edge, or trying to figure out who or what lives on earth's underbelly. But when you learn that the earth is round, you see that the previous mysteries are not solved; instead, they're rendered irrelevant.

During the first decades of the twentieth century, Albert Einstein made two deep discoveries. Each caused a radical upheaval in our understanding of space and time. Einstein dismantled the rigid, absolute structures that Newton had erected, and built his own tower, synthesizing space and time in a manner that was completely unanticipated. When he was done, time had become so enmeshed with space that the reality of one could no longer be pondered separately from the other. And so, by the third decade of the twentieth century the question of the corporeality of space was outmoded; its Einsteinian reframing, as we'll talk about shortly, became: Is *spacetime* a something? With that seemingly slight modification, our understanding of reality's arena was transformed.

Is Empty Space Empty?

Light was the primary actor in the relativity drama written by Einstein in the early years of the twentieth century. And it was the work of James Clerk Maxwell that set the stage for Einstein's insights. In the mid-1800s, Maxwell discovered four powerful equations that, for the first